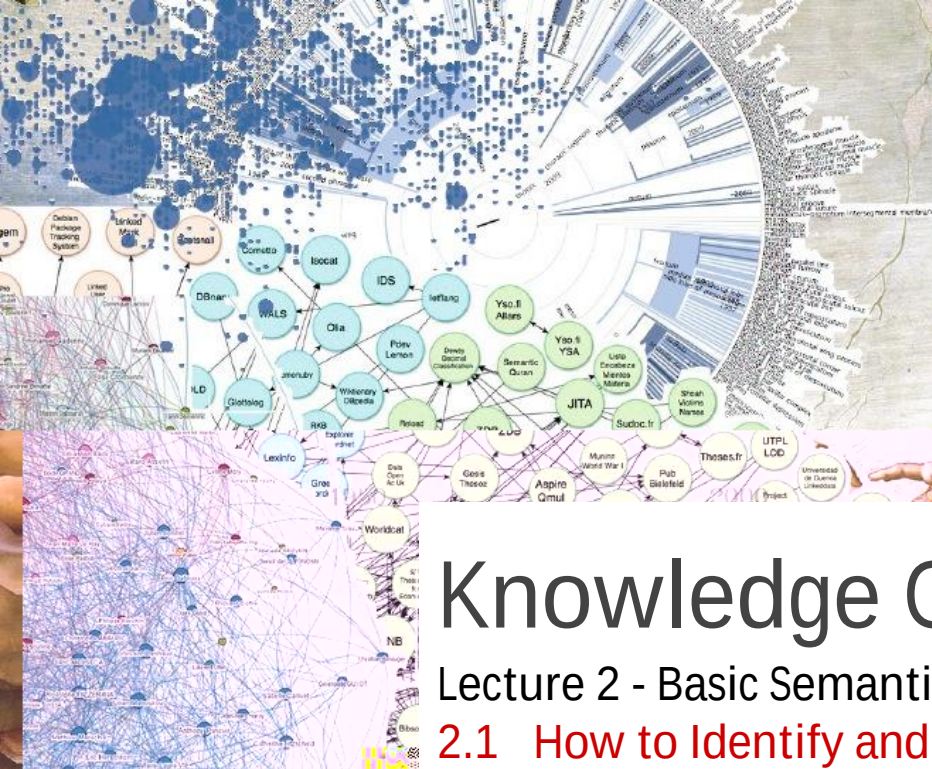




Knowledge Graphs

Lecture 2 - Basic Semantic Technologies

2.1 How to Identify and Access Things

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Autumn 2020



Knowledge Graphs

Lecture 2: Basic Semantic Technologies

2.1 How to Identify and Access Things

2.2 How to Represent Simple Facts with RDF

2.3 RDF Turtle Serialization

2.4 RDF Complex Data Structures

2.5 Model Building with RDFS

2.6 Logical Inference with RDF(S)

Excursion 1: RDFa - RDF and the Web

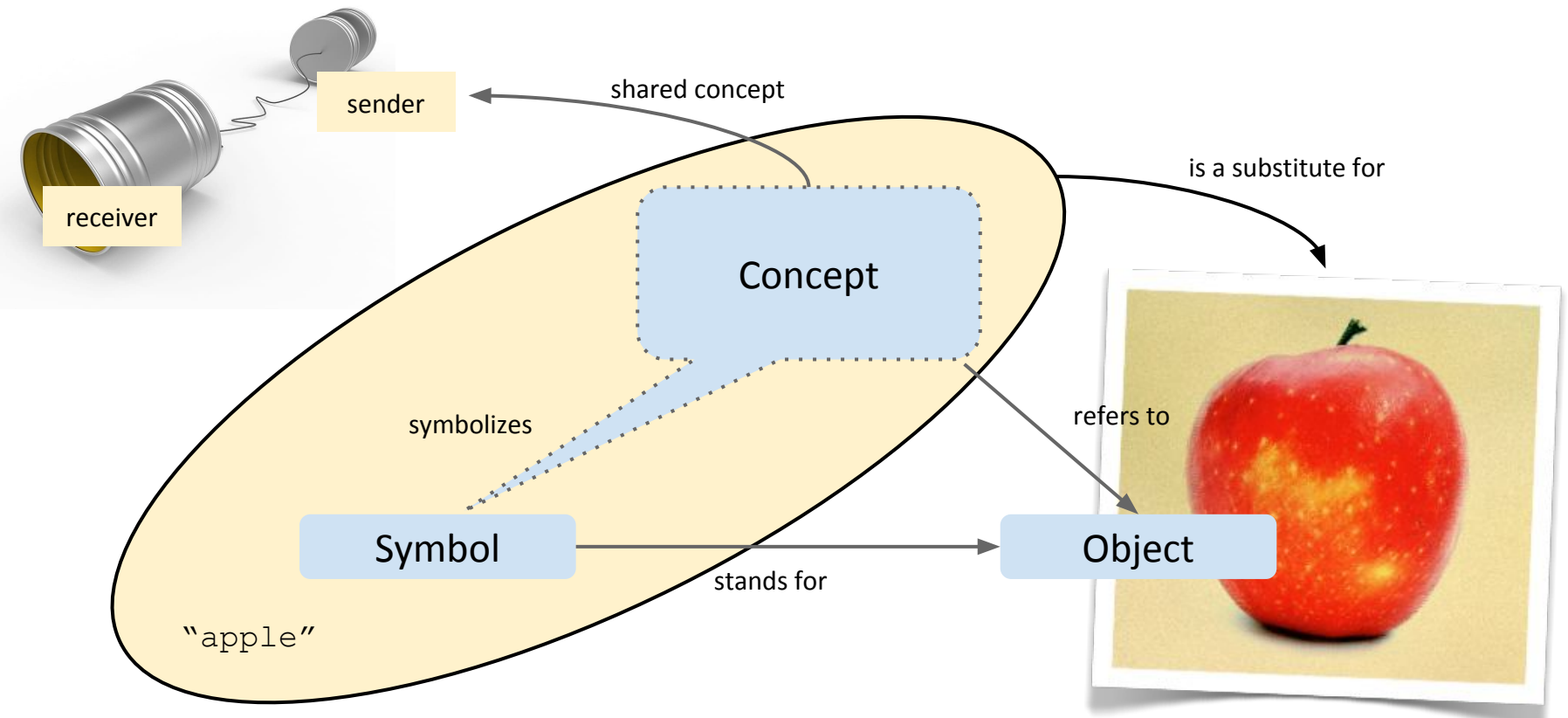
What is this?





Ceci n'est pas une pomme.

Semiotic Triangle



[1,2]

Uniform Resource Identifier

- A Uniform Resource Identifier (URI) defines a simple and extensible schema for worldwide unique identification of abstract or physical resources (RFC 3986).

https://www

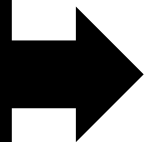
Uniform Resource Identifier

- A Uniform Resource Identifier (URI) defines a simple and extensible schema for worldwide unique identification of abstract or physical resources (RFC 3986).
- A Resource can be every object with a clear identity (according to the context of the application)
 - as e.g., web pages, books, locations, persons, relations among objects, abstract concepts, etc.
- URI concept is already established in various domains, as e.g.,
 - the Web (URL),
 - Books and publications (ISBN, ISSN),
 - Digital Object Identifier (DOI).

URL

identify
what exists
on the web

<https://www.mywebsite.org/>

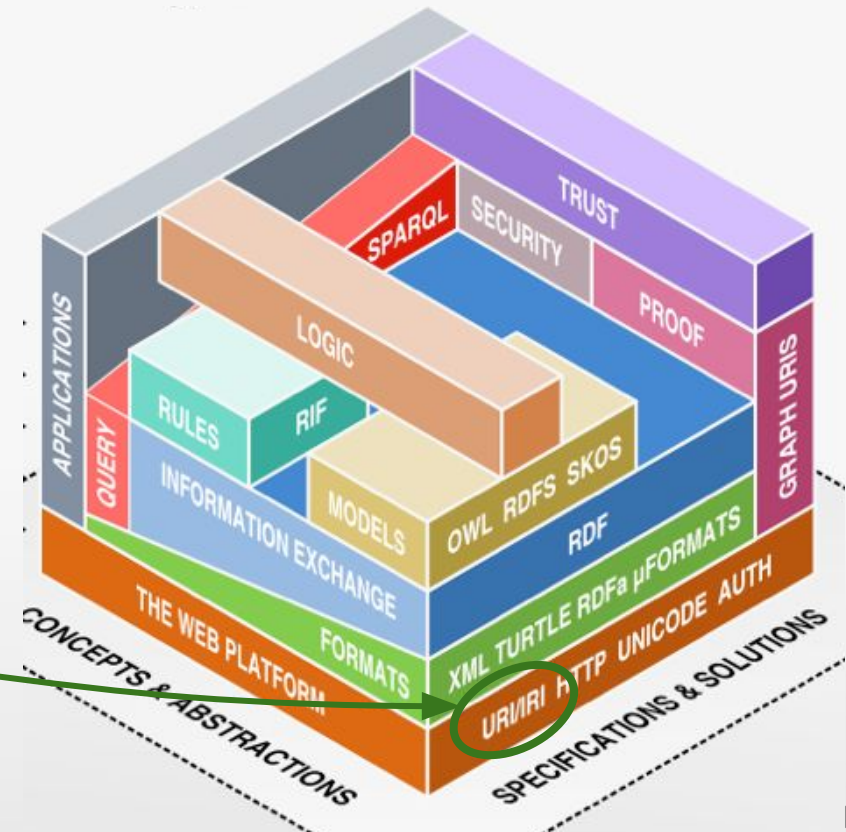
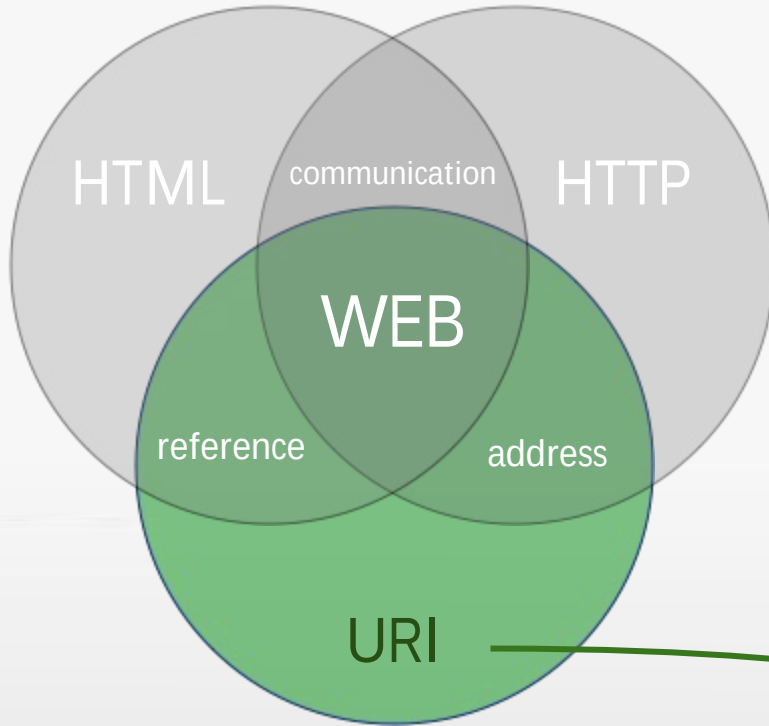


URI

identify
on the web
what exists

https://fiz-karlsruhe.de/harald_foaf.rdf#me

Basic Architecture of the Web of Data



URI

Resource

<http://en.wikipedia.org/wiki/Earth/>

identifies

Designatum

describes

represents (stands for)

Representation

Designator

Metadata:

Content-type: text/html

Data:

```
<!DOCTYPE html>
```

```
<html class="client-nojs" lang="en" dir="ltr">
```

```
<head>
```

```
<meta charset="UTF-8"/>
```

```
<title>Earth - Wikipedia</title>
```

```
...
```

```
</html>
```

defines

The Earth is described
(designated) by a web page.

What if a URI does not exist (yet)?

- Define a URI by yourself:
 - avoid overlaps → use your own website as namespace
 - enable documentation at the same place → HTTP Content Negotiation
- Use separate URIs for the resource (*Designatum*) and its documentation (*Designator*) via
 - HTTP Content Negotiation and/or
 - URI references (e.g. via "#" fragment identifier)

HTTP Content Negotiation

Let's try an example:

- I want to have information about the Earth from DBpedia.

1 HTTP GET request
Accept Header: text/html

<http://dbpedia.org/resource/Earth>

URI represents *Designatum*

2 HTTP/2 303 See Other

<http://dbpedia.org/page/Earth>

3 URI represents *Designator*

HTTP GET request
Accept Header: text/html

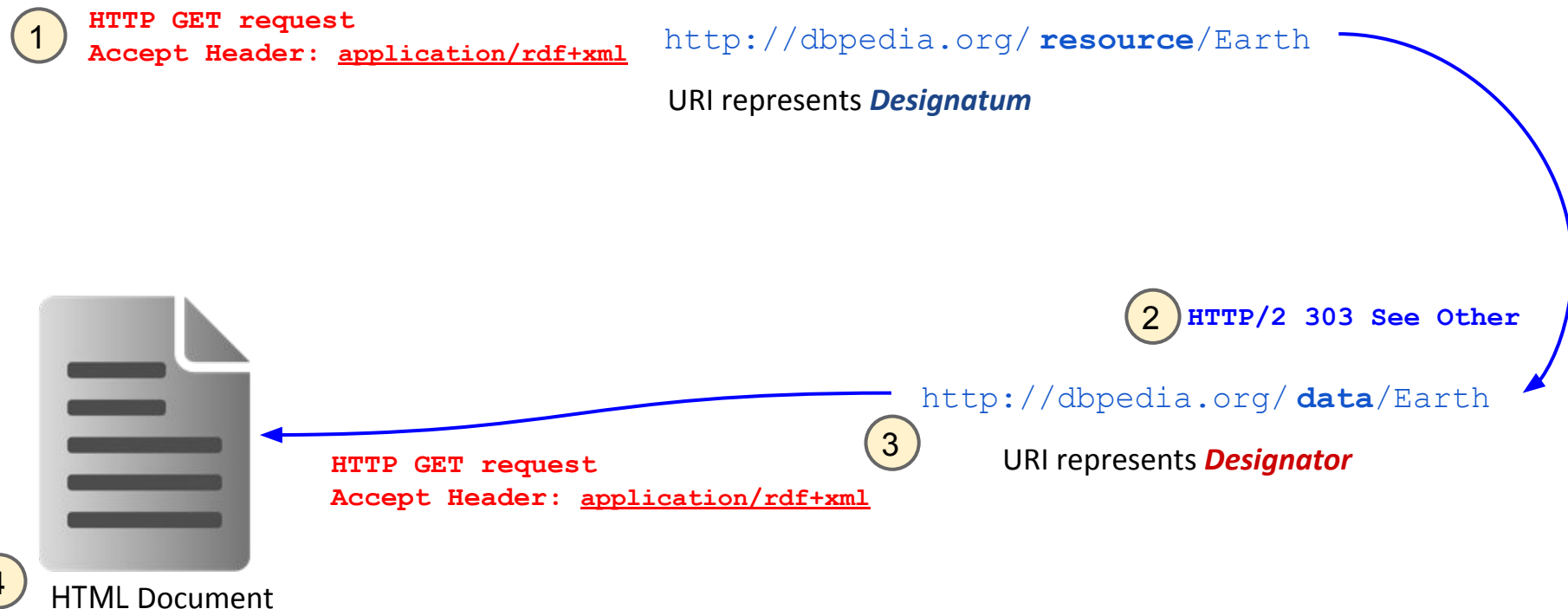


4 HTML Document

HTTP Content Negotiation

Let's try another example:

- I want to have **machine readable information** about the Earth from DBpedia.



HTTP Content Negotiation

Let's try it ourselves:

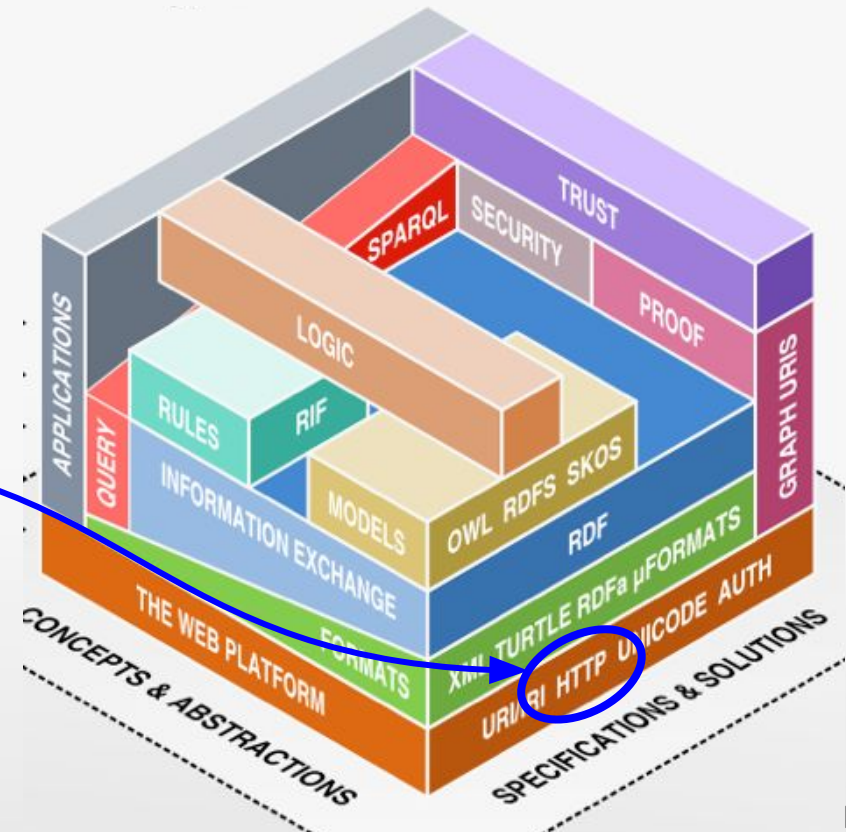
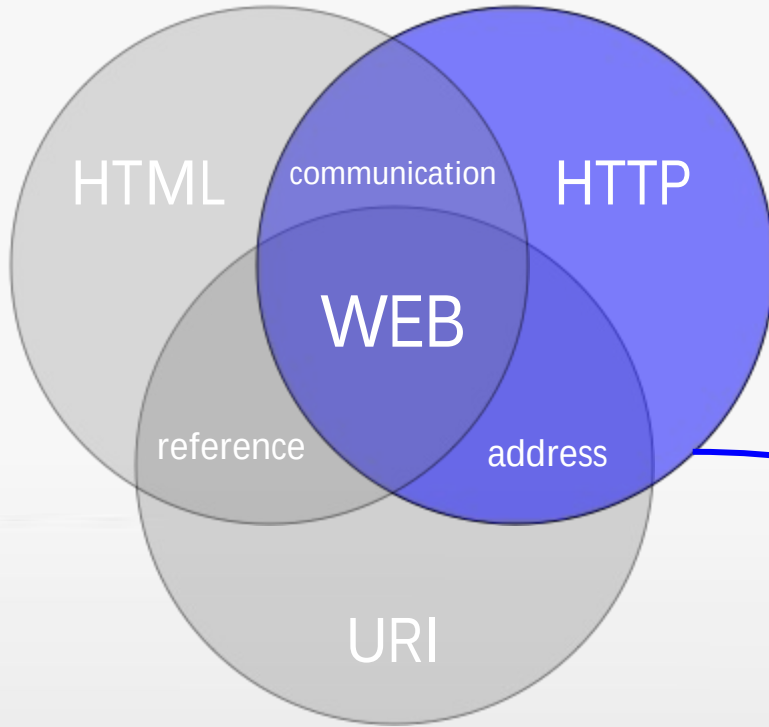
- I want to have information about the Earth from DBpedia.

```
curl -L -H "Accept: text/html" http://dbpedia.org/resource/Earth
```

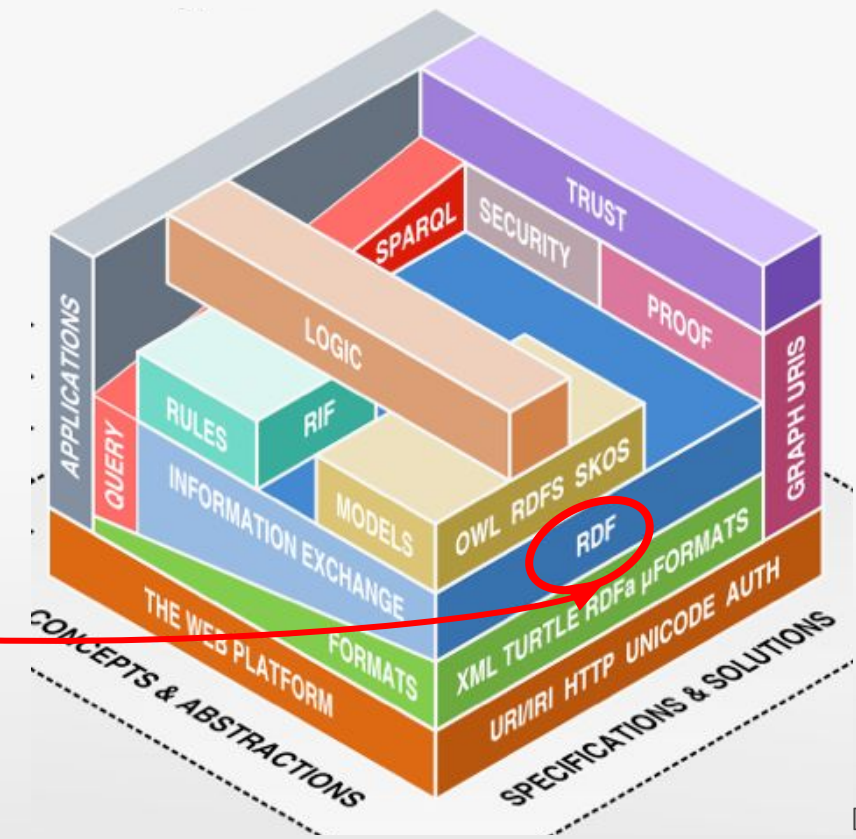
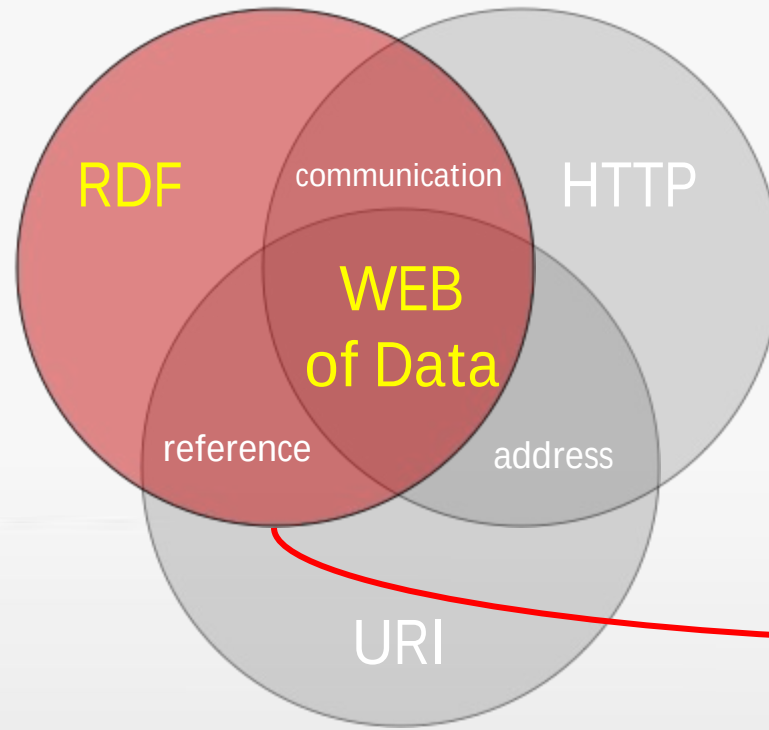
- I want to have machine readable information about the Earth from DBpedia.

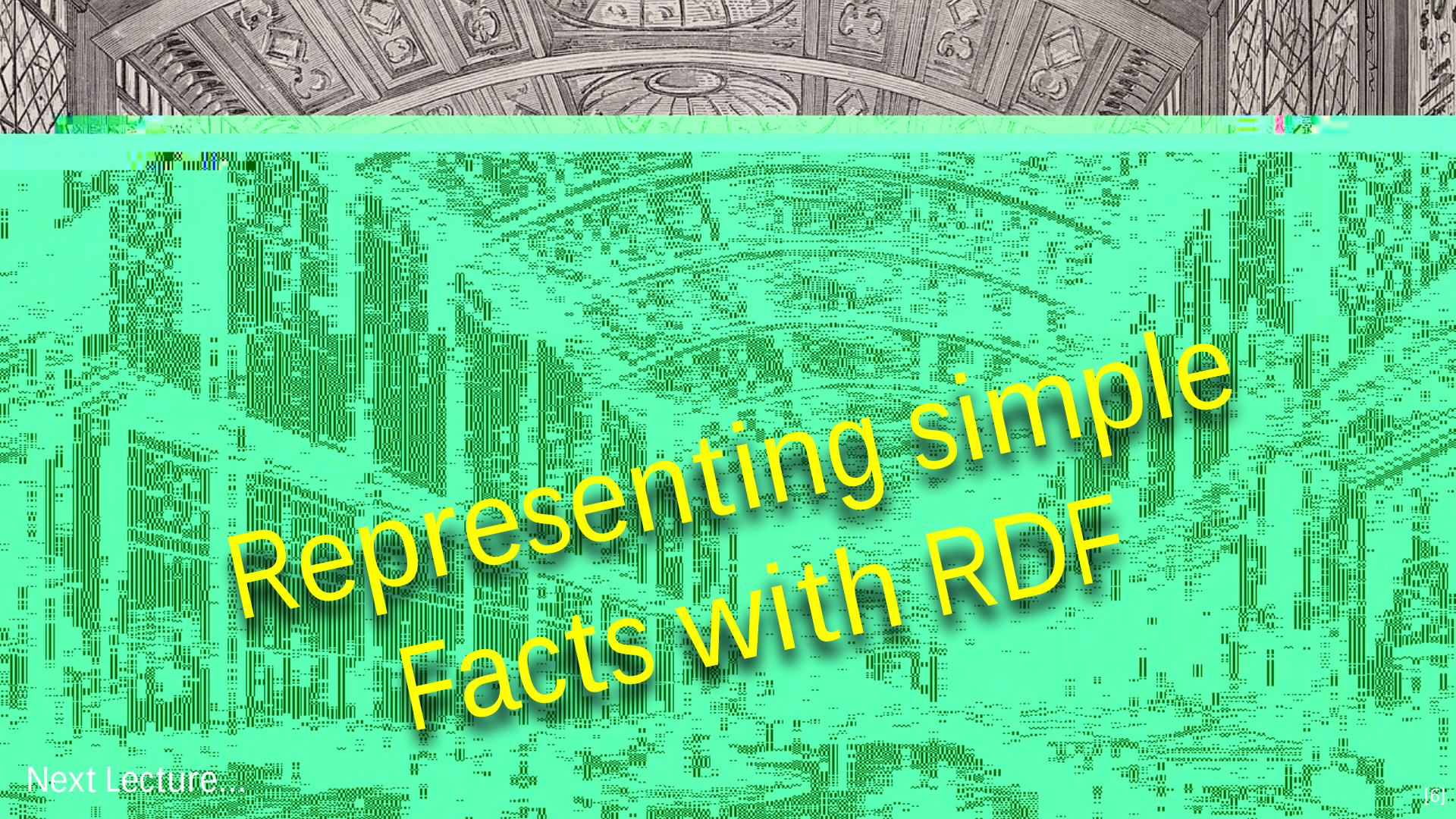
```
curl -L -H "Accept: application/rdf+xml" http://dbpedia.org/resource/Earth
```

Basic Architecture of the Web of Data



Basic Architecture of the Web of Data





Representing simple Facts with RDF

Next Lecture...

Picture References:

- [1] Johannes Osterhoff, *Ceci n'est pas une pomme*, in Harald Sack, *Semantic Web Technologien*, HPI lecture 2010/11
- [2] Phone Conversation Indicating Chit Chat And Discussion, CCO
<https://www.pigsels.com/en/public-domain-photo-jrdox>
- [3] The new https protocol, Fabio Lenari, [CC-BY-SA]
<https://commons.wikimedia.org/wiki/File:Internet2.jpg>
- [4] "The Blue Marble" is a famous photograph of the Earth taken on December 7, 1972, by the crew of the Apollo 17 spacecraft en route to the Moon at a distance of about 29,000 kilometres (18,000 mi). It shows Africa, Antarctica, and the Arabian Peninsula, NASA, public domain
https://commons.wikimedia.org/wiki/File:The_Earth_seen_from_Apollo_17.jpg
- [5] Benjamin Nowack, *The Semantic Web - Not a Piece of cake ...*, at bnode.org, 2009-07-08 , [CC BY 3.0]
<http://bnode.org/blog/2009/07/08/the-semantic-web-not-a-piece-of-cake>
- [6] Trichon, La galerie du travail manuel, dans le palais du Champ-de-Mars.(1879), Brown University Library [Public domain]
https://commons.wikimedia.org/wiki/File:La_galerie_du_travail_manuel,_dans_le_palais_du_Champ-de-Mars.jpg